



CHAPTER 8

Long-Term Operating Assets

Property, Plant & Equipment · Intangible Assets · Asset-Efficiency Analysis

Introduction to Financial Accounting | KAIST College of Business

Study Plan

LECTURE TOPICS

1	Introduction — the three PPE questions	8.0 – 8.1
2	Determining costs to capitalize	8.2
3	Depreciation — concept, methods, tax	8.3
4	Asset sales & impairments	8.4
5	Intangibles — nature, R&D, capitalization	8.5 – 8.6
6	Amortization, impairment & goodwill	8.7 – 8.8
7	Analysis — PPE turnover & percent depreciated	8.9
8	Discussion questions & judgment calls	8.10

Seven concept quizzes are embedded throughout, with answers revealed on the following slide.

PART 1

Property, Plant & Equipment

Capitalize · Allocate · Write down

One Aircraft, Three Income Statements

THE SETUP

Before any rules: an airline buys one aircraft for \$100m. Three accountants book it three ways. Who is right?

Accountant A

\$0

capitalized

Expense everything

Books the full \$100m as an expense this year.

Year-1 profit:
crushed.

Accountant B

\$100m

capitalized

Capitalize everything

Books \$100m as an asset, spreads it over 20 years.

Year-1 profit:
barely dented.

Accountant C

\$108m

capitalized

Capitalize + fit-out

Adds \$8m of cabin fit-out and delivery costs to the asset.

Year-1 profit:
highest of the three.

All three are defensible under some reading of the facts — and each produces a different net income. That gap is what this chapter is about: not arithmetic, but judgment.

The Three Questions That Run the Chapter

SECTION 8.0

PPE is usually the largest and longest-lived asset a company owns. Every topic ahead answers one of three questions.

1

Capitalize or Expense?

Which costs go on the balance sheet as assets, and which hit the income statement immediately?

2

How to Allocate?

How do we spread the capitalized cost across the periods that actually benefit from the asset?

3

When to Write Down?

How do we report asset sales, and declines in an asset's recoverable value?

ROADMAP

Part 1 answers all three questions for tangible assets.

Part 2 applies the same logic to intangibles — with their own twists (R&D, goodwill).

Part 3 turns the numbers into ratios that reveal how efficiently assets generate sales.

Tangible vs. Intangible Long-Term Operating Assets

SECTION 8.1

Long-term operating assets are used in operations for more than one period — they are not held for resale.

Tangible Assets

- **Have physical substance**
- Land and buildings
- Machinery and equipment
- Vehicles, fixtures, leasehold improvements

Intangible Assets

- **Lack physical substance, but grant rights**
- Patents, trademarks, copyrights
- Franchise rights, licence agreements
- Goodwill

The Two Traits They Share

- Used to generate revenue — not resold in the ordinary course of business.
- Benefit multiple accounting periods — so cost must be allocated over time, not expensed at once.

Purpose Drives Classification

SECTION 8.1

The same physical truck can be three different balance-sheet accounts. It depends entirely on why the company holds it.

Inventory

Held for resale in the ordinary course of business.

Example A truck on a dealer's lot.

PP&E

Used in operations for more than one period to generate revenue.

Example A delivery company's truck.

Held for Sale

No longer used in operations, but not normal sales inventory either.

Example A retired truck awaiting resale.

Ask: is it for resale, for use, or retired-and-for-sale? The purpose — not the object — decides the account.

Why it matters: the classification drives where the cost goes — cost of goods sold, depreciation, or neither.

Capitalize or Expense?

SECTION 8.2

The single most consequential judgment in this chapter — it shifts income between periods without changing a single cash flow.

Capitalize

- Record the cost as an asset on the balance sheet.
- Depreciated or amortized later, over the periods it benefits.
- **Current-year income: protected.**

Expense

- Record the cost immediately on the income statement.
- Nothing carries forward to future periods.
- **Current-year income: reduced right away.**

TWO REQUIREMENTS TO CAPITALIZE

1

Owned or Controlled The asset is owned or controlled by the company.

2

Future Benefit The asset is expected to provide future economic benefits.

Capital expenditure (asset) vs. revenue expenditure (period expense). Same cash out the door — very different reported income.

What Costs Get Capitalized?

SECTION 8.2

Capitalize every normal cost required to acquire the asset and prepare it for its intended use.

Include in Cost

- Purchase price
- Shipping, taxes, legal fees
- Installation, setup, calibration
- End-of-life legal obligations (site restoration) — capitalized AND booked as a liability

Self-Constructed Assets

- Materials, labour and reasonable overhead
- **Capitalized interest**
- Interest incurred during the construction period is added to the asset, not expensed

Two Guardrails

- **Direct linkage**
- Only costs directly tied to future benefits qualify. Incidental costs are expensed.
- **Future-benefit ceiling**
- Capitalized cost cannot exceed expected future cash inflows.

The ceiling in one line: a \$200 asset on the books should generate at least \$200 of future cash flows. If it cannot, the excess is a loss today.

Watch the boundary: routine repairs keep the asset in its existing condition (expense); improvements extend life or capacity (capitalize).

The Capitalize-or-Expense Decision, as a Flow

DECISION TOOL

Four questions, asked in order. The first 'no' ends the process.



Four yeses → capitalize the full cost, then allocate it over the periods that benefit. Everything else is a period expense.

Depreciation — the Concept

SECTION 8.3

The systematic allocation of a long-term asset's cost to each period that benefits from its use.

Income Statement

- **Depreciation Expense**
- Reduces net income each period the asset is in service.

Balance Sheet

- **Accumulated Depreciation**
- A contra-asset account ("XA") — the offsetting credit, netted against asset cost.

$$\text{Book value (carrying value)} = \text{Asset cost} - \text{Accumulated depreciation}$$

Read that carefully: book value is a running total of cost not yet allocated. It is not an appraisal, not a market price, and not an estimate of what the asset would fetch.

Long-lived assets benefit many periods, so their cost cannot be matched to one period's revenue. Allocation is the only defensible answer.

A Critical Caveat: Book Value $\neq$ Fair Value

SECTION 8.3

Depreciation is cost allocation. It is not, and never claims to be, a measure of decline in market value.

IN ANY GIVEN PERIOD, THE ASSET'S FAIR VALUE MAY...



Decline more

than the depreciation recorded



Decline less

than the depreciation recorded



Even increase

e.g. appreciating land and buildings

The consequence students forget

- A fully-depreciated machine has a book value of zero — and may still be running profitably for another decade.
- A building carried at cost less 30 years of depreciation may be worth far more than its carrying amount.

Depreciation follows a formula. The market does not.

Comparing Methods — Straight-Line vs. SYD

SECTION 8.3

Dehning Co. delivery truck: cost \$120,000 · useful life 5 years · residual value \$20,000 · depreciable base \$100,000.

Method 1 — Straight-Line

Same expense every year · rate = $1/5 = 20\%$

Yr	Rate	Dep. Exp.	Book Value
1	20%	\$20,000	\$100,000
2	20%	\$20,000	\$80,000
3	20%	\$20,000	\$60,000
4	20%	\$20,000	\$40,000
5	20%	\$20,000	\$20,000
Tot	100%	\$100,000	—

Method 2 — Sum-of-Years'-Digits

Front-loaded · denominator = $1+2+3+4+5 = 15$

Yr	Rate	Dep. Exp.	Book Value
1	5/15	\$33,333	\$86,667
2	4/15	\$26,667	\$60,000
3	3/15	\$20,000	\$40,000
4	2/15	\$13,333	\$26,667
5	1/15	\$6,667	\$20,000
Tot	100%	\$100,000	—

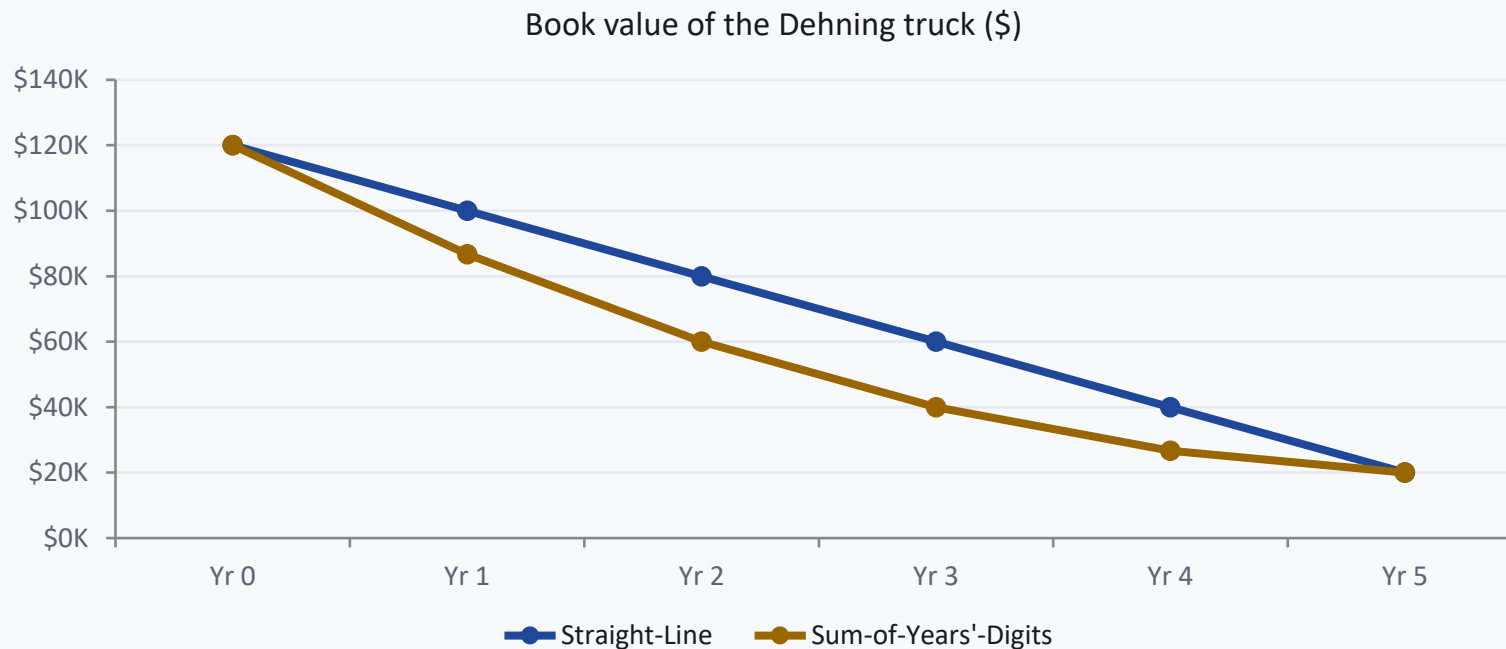
Key takeaway: identical total depreciation (\$100,000) and identical final book value (\$20,000). The methods differ only in **timing** — SYD loads expense into the early years.

Which method a manager picks changes reported profit for five years and total profit not at all.

The Same Destination, Two Different Roads

SECTION 8.3

The tables on the previous slide are one picture: both methods start at \$120,000 and land on \$20,000. Only the path differs.



Read the gap

At the end of Year 1 the SYD truck is already \$13,333 cheaper on the balance sheet — and \$13,333 of profit has vanished from the income statement.

Then it reverses

By Year 4–5 SYD depreciation is smaller than straight-line, so the SYD firm now reports the higher profit. Nothing economic changed.

Any analyst comparing two firms' margins must first ask whether they are on the same depreciation road.

Depreciation for Tax Purposes

SECTION 8.3

Most companies legally keep two sets of numbers: straight-line for the financial statements, accelerated for the tax return.



The benefit — tax deferral

Paying less tax now and more later is an interest-free loan from the government. A dollar saved today is worth more than a dollar paid in five years.

The consequence

Book–tax differences create deferred tax liabilities on the balance sheet — the tax you have postponed, not escaped.

The identity to remember: total depreciation and total tax over the asset's life are identical under either method. Only the **timing** of the cash tax shifts.

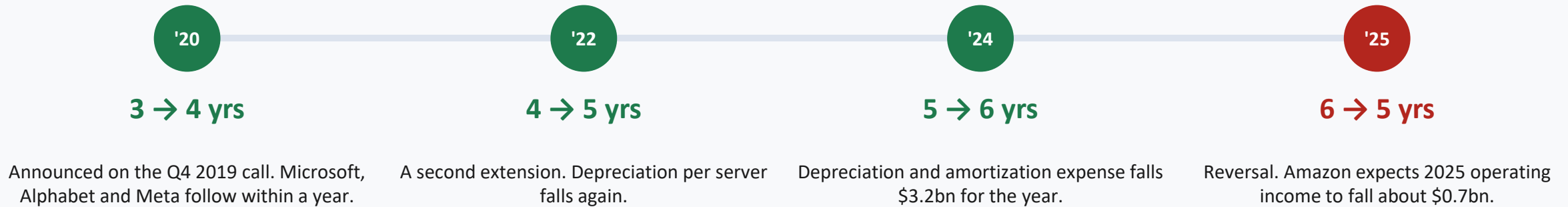
In the U.S. the accelerated tax system is MACRS. Korea's tax code permits 정률법 (declining balance) alongside 정액법 — the same deferral logic applies.

The Quietest Earnings Lever in Accounting

JUDGMENT

Useful life is an estimate. Change the estimate and profit moves — no transaction, no cash, no restatement.

AMAZON'S SERVER USEFUL LIFE, AS DISCLOSED IN ITS OWN FILINGS



The same study, the opposite direction

In that same Q4 2024 review Amazon extended heavy equipment from 10 to 13 years — lifting 2025 operating income by about \$0.9bn, while the server change was cutting it by \$0.7bn.

Why servers went back down

Amazon cited the increased pace of technology development, particularly in AI and machine learning. It also booked roughly \$920m of accelerated depreciation on equipment retired early.

All of it is legal: a change in accounting estimate is applied prospectively — no prior year restated, no rule broken, and the disclosure sits in a footnote most readers never reach.

Percent depreciated (Section 8.9) is one of the few ratios that quietly flags an asset base that keeps getting younger.

Gains & Losses on Asset Sales

SECTION 8.4

On disposal, BOTH the original cost and the accumulated depreciation leave the balance sheet.

$$\text{Gain or Loss} = \text{Proceeds from Sale} - \text{Book Value of the Asset Sold}$$

Gain (Proceeds > Book Value)

Recognized as income on the income statement.

Loss (Proceeds < Book Value)

Recognized as a loss on the income statement.

Illustration — the Dehning delivery truck, sold at the end of Year 4

Straight-line: cost \$120,000, accumulated depreciation after 4 years \$80,000, so book value is \$40,000. Dehning sells the truck for \$42,000 cash.

Dr	Cash	\$42,000	Dr	Accumulated Depreciation	\$80,000	
	Cr	Truck	\$120,000	Cr	Gain on Sale	\$2,000

Proceeds \$42,000 – book value \$40,000 = **\$2,000 gain** — which is really just an admission that 5-year straight-line depreciated the truck slightly too fast.

Asset Impairments — the Two-Step Test

SECTION 8.4

Write-ups are prohibited. Write-downs are required — but only when the decline is other-than-temporary.

Step 1 · Recoverability test — is it impaired?

Sum of UNDISCOUNTED future cash flows vs. net book value

Cash flows > NBV → no impairment. Stop.

Cash flows < NBV → impaired. Go to Step 2.

Note the asymmetry: an undiscounted comparison is a deliberately low hurdle. GAAP does not want firms writing down at every wobble.

Step 2 · Measurement — by how much?

Write down to fair value — the DISCOUNTED present value of those flows

The written-down amount is permanently removed from the asset's cost.

Going forward, depreciation is computed on the lower base — so future depreciation expense falls and future profits look better.

The line to memorise: undiscounted cash flows decide *whether* to impair; discounted cash flows decide *how much*.

Because a write-down lowers all future depreciation, a large impairment today mechanically raises reported profit for years afterwards. Hold that thought.

Impairment — Where Judgment Becomes Risk

SECTION 8.4

Impairment timing and amount both rest on management's own cash-flow forecasts. That is the whole problem.

Abuse 1 — Insufficient write-down

Management delays or under-books the loss.

Overstates current income; understates future income. The pain is postponed, not avoided.

Abuse 2 — Aggressive write-down ("big bath")

Management purges every doubtful cost into one already-bad year.

Future depreciation is lower, so future earnings and future ROA look better — often just as a new CEO arrives.

Why impairments hide inside restructuring charges

Plant closures and product-line exits reduce expected cash flows, which triggers the impairment test — so the write-down arrives bundled with severance and exit costs in a single restructuring line. The bundling is convenient: it obscures how much of the charge was a genuine operating decision and how much was discretionary.

Keep this straight: an impairment is a non-cash charge — but the cash was already spent at acquisition. The write-down simply admits it will not be recovered.

K-IFRS vs. U.S. GAAP — Two Answers to One Question

COMPARATIVE

Everything so far has been the U.S. GAAP answer. Korean listed companies report under K-IFRS, and on this topic the two diverge sharply.

U.S. GAAP

- Cost model only.
- PPE stays at cost less accumulated depreciation.
- Upward revaluation is prohibited.
- Impairment reversals are prohibited.

K-IFRS / IFRS

- Cost model or revaluation model (기업회계기준서 제1016호).
- PPE may be carried at fair value, revalued regularly.
- Increases go to OCI as 재평가잉여금.
- Impairment reversals are permitted.

Why it mattered in Korea

When Korean companies adopted K-IFRS around 2011, a number of them elected the revaluation model for land and buildings held at decades-old historical cost. Revaluing lifted total assets and equity — improving the 부채비율 that lenders and regulators watch — without any operating improvement at all.

Same asset, same economy, two carrying amounts. Cross-border ratio comparison starts by asking which model the firm elected.

PART 2

Intangible Assets

Same three questions · harder answers

Why Isn't the Swoosh on Nike's Balance Sheet?

THE PUZZLE

One of the most valuable trademarks on earth is carried at essentially nothing. This is not an oversight — it is the rule.

Company A — builds the brand

Spends 40 years and billions on advertising, design and sponsorship to create a globally recognised mark.

Accounting treatment: every won of it is an expense. The trademark never appears as an asset.

Company B — buys the brand

Acquires exactly the same mark from Company A for its market price.

Accounting treatment: the full purchase price is capitalised as an intangible asset on the balance sheet.

Identical asset. Identical economics. Completely different balance sheets.

Hold the question: if Company B now reports a lower ROA than Company A purely because its assets are bigger, which firm is actually performing better? Part 3 comes back to this.

The Nature of Intangible Assets

SECTION 8.5

Assets that lack physical substance but confer specific property or legal rights — patents, trademarks, franchise rights, goodwill.

Challenge 1 — benefits are uncertain

Future benefits of an intangible are often impossible to quantify with confidence. Will the patent produce a product? Will the brand hold its price premium?

Challenge 2 — useful life is hard to estimate

Legal life and economic life rarely coincide, and some intangibles have no foreseeable limit at all — an indefinite life.

The same three questions from Part 1

Capitalize or expense? · How do we allocate the cost? · When do we write it down?

But for intangibles the answers hinge on one extra fact that never mattered for PPE: was the asset purchased, or was it built in-house?

Depreciation becomes amortization. Salvage value is usually assumed to be zero. The method is almost always straight-line.

Purchased vs. Internally Developed

SECTION 8.5

This single distinction drives almost every intangible-asset answer in U.S. GAAP.

Purchased from another party

- **CAPITALIZE** the purchase price.
- Recorded as an intangible asset, then amortized over its useful life.

Developed internally

- **EXPENSE** as incurred.
- R&D and internal brand-building go straight to the income statement. Narrow exception: legal and registration fees to secure the right.

Research & development under U.S. GAAP

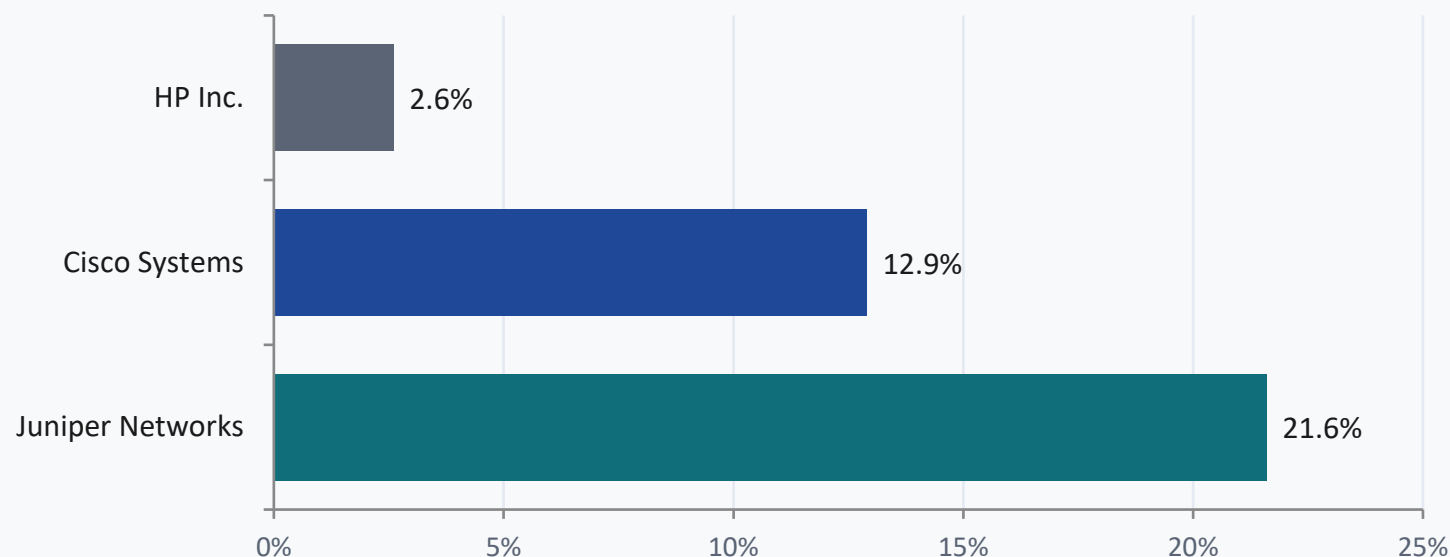
ALL R&D costs are expensed as incurred — future benefits are judged too uncertain to capitalize. R&D plant assets with an alternative future use are capitalized and depreciated; assets used solely for R&D with no alternative use are expensed.

The asymmetry: a firm that built its brand reports a far smaller intangible balance than one that bought the identical brand. This is the most-criticised feature of U.S. GAAP in this area — and IFRS answers it differently.

How Much Gets Expensed? R&D Intensity

SECTION 8.5

R&D as a percentage of sales. Every won of this is expensed the year it is spent — none of it becomes an asset.



The pattern

High-tech firms routinely spend 10–20% of revenue on R&D. Consumer-hardware assemblers spend a fraction of that.

The distortion

Because GAAP expenses all of it, R&D-heavy firms carry understated assets and equity — which mechanically inflates ROA and ROE.

Juniper's asset base is understated by roughly a fifth of revenue every year. Comparing its ROA to HP's without adjusting is comparing two different accounting worlds.

Source: company filings, fiscal 2020. Cisco 12.9%, Juniper 21.6%, HP 2.6% — all three were stable across 2018–2020.

R&D Expensing — the Second-Order Effects

SECTION 8.5

An accounting rule designed for conservatism ends up shaping how managers actually spend money.

Assets understated

Decades of R&D that produced real, durable know-how appear nowhere on the balance sheet.

Book value drifts far below any sensible economic value.

Ratios inflated

ROA and ROE rise because the denominator is too small — not because the business is better.

Pharma and software look spuriously profitable next to manufacturers.

Behaviour distorted

Because R&D hits earnings immediately, cutting it is the fastest way to hit a quarterly target.

Research documents exactly this: managers under earnings pressure cut R&D.

The myopia problem

Spending that should improve future performance depresses current earnings, so a short-horizon manager cuts it — and the accounting rule, not the economics, made that trade-off attractive. Some firms decline to disclose R&D spending at all, likely to keep the scale of their investment away from competitors.

Analysts often capitalize R&D themselves to undo the distortion — a reminder that reported numbers are a starting point, not an answer.

The Korean Version — 개발비 자산화

COMPARATIVE

IFRS splits R&D in two: research is expensed, development may be capitalized once six criteria are met.

Research → expense

Investigation undertaken to gain new scientific or technical knowledge, with no certainty of a product.

Always expensed as incurred, exactly as under U.S. GAAP.

Development → may be capitalized

Applying research findings to a planned product or process. Capitalize only if all six IAS 38 criteria are met — technical feasibility, intention and ability to complete, probable future benefits, adequate resources, reliable measurement.

WHERE KOREA DREW THE LINE — 금융위·금감원 「제약·바이오 기업의 연구개발비 회계처리 관련 감독지침」, 2018년 9월

신약 (new drug)

임상 3상 개시 승인 이후

Before Phase 3 approval, spending goes to 경상연구개발비 — expensed in the period.

바이오시밀러

임상 1상부터

A known reference product means feasibility is demonstrable far earlier.

제네릭·진단시약

별도 단계 설정

The guidance also fixes capitalization stages for generics and diagnostic reagents.

U.S. GAAP removes the judgment and accepts the distortion. IFRS keeps the judgment — and Korea discovered it then needed a rulebook to make the judgment comparable.

Patents, Copyrights, Trademarks & Franchise Rights

SECTION 8.6

Four rights, four very different lives — and life is what drives the accounting.

Intangible	What the right actually is	Useful life
Patent	Exclusive right to produce a product or use a technology, granted by government	Legal life 20 yrs; useful life often much shorter
Copyright	Exclusive right of an author, composer or artist over creative works	Life of the creator + 70 years
Trademark	Registered name, logo, slogan or design	Indefinite — renewable forever
Franchise Right	Contractual right to operate a business in a defined area for a defined period	The contractual period

The rule that follows from the last column

A trademark's indefinite life means it is NOT amortized. Instead it is tested for impairment annually — exactly like goodwill. Definite-life intangibles are amortized; indefinite-life intangibles are tested.

"Indefinite" does not mean infinite. It means the period of benefit cannot be reliably estimated.

Capitalization Rules — Three Cases, Three Answers

SECTION 8.6

The same trademark or patent gets three different treatments depending on how the company came by it.

1

Purchased patent

The purchase price is capitalized as an intangible asset.

Amortized over the shorter of legal and useful life.

2

Internally developed patent

R&D costs are expensed.

ONLY the legal and registration fees are capitalized — often a trivial amount next to the research behind it.

3

Internally developed trademark

Advertising and development costs are expensed.

The trademark itself never reaches the balance sheet at all.

This is why the world's most valuable trademarks are not significant assets on their owners' books.

Back to the puzzle: Case 3 is Company A from the opening slide of Part 2. Case 1 is Company B.

Amortization & Impairment of Intangibles

SECTION 8.7

Amortization is the intangible-asset equivalent of depreciation: systematic allocation of cost over useful life, almost always straight-line.

Definite lives → amortize

- Patents, copyrights, franchise rights, licences
- Straight-line over the useful life
- Salvage value normally assumed to be zero

Indefinite lives → do NOT amortize

- Many trademarks, and goodwill
- No amortization at all
- Tested annually for impairment instead

When legal life and useful life differ, amortize over the SHORTER of the two.

Worked in one line: a patent with a 12-year remaining legal life but a 5-year expected useful life is amortized over 5 years. Benefits drive the period, not legal rights.

The economic logic is identical to PPE. Only the vocabulary changes — and salvage value quietly disappears.

Illustration — the Landsman Patent

SECTION 8.7

Landsman spends \$100,000 in early 2026 to purchase a patent. Remaining legal life 12 years; estimated useful life 5 years.

(1) Capitalize the purchase — 2026

Dr Patent (asset)	\$100,000
Cr Cash	\$100,000

(2) Annual amortization — end of 2026

Dr Amortization Expense	\$20,000
Cr Patent	\$20,000

$$\$100,000 \div 5 \text{ years (useful life)} = \$20,000 \text{ per year}$$

The point of the question: amortize over the 5-year useful life, NOT the 12-year legal life. Using 12 years gives \$8,333 and understates expense by 60%.

Note the credit: amortization is credited directly to the Patent account. Intangibles usually carry no separate accumulated-amortization contra account, unlike PPE.

Goodwill

SECTION 8.8

The one intangible that cannot be bought on its own, cannot be built, and cannot be separated from the business it came with.

$$\text{Goodwill} = \text{Purchase Price} - \text{Fair Value of Identifiable Net Assets Acquired}$$

Arises only in an acquisition

Recognized only when one company acquires another and pays more than the fair value of the identifiable net assets it receives.

Internally generated goodwill is NEVER recorded — no matter how strong the reputation.

What the buyer is actually paying for

Brand strength, assembled workforce, customer relationships, expected synergies — real sources of value that cannot be separately identified and measured.

Goodwill is the accounting residual for all of it.

Illustration — Procter & Gamble / Gillette

\$53.4bn

paid for Gillette in 2006

\$35.3bn

recorded as goodwill

\$39.9bn

still on the books in 2020

Goodwill Impairment — the One-Way Door

SECTION 8.8

Goodwill is treated as having an indefinite life, so it is never amortized. It sits at its original amount until the day it is written down.

Not amortized

Indefinite life, so no periodic charge. Goodwill can sit unchanged on a balance sheet for decades.

Tested annually

If the fair value of the acquired reporting unit falls below its carrying amount, goodwill is written down to an imputed value.

No recoveries

Once impaired, always impaired. Write-ups are prohibited under U.S. GAAP — and under IFRS as well for goodwill.

Case — Hewlett-Packard / Autonomy, 2012

\$8.8bn

write-down taken

HP acquired Autonomy in 2011 for roughly \$11bn and wrote off \$8.8bn barely a year later — attributing more than \$5bn of the charge to accounting improprieties at the acquired company, the rest to overpayment and deteriorating performance.

A goodwill write-down of that scale is management publicly conceding that the price paid was wrong.

Large, non-cash, non-recurring — and almost always a signal about the past, not the future.

PART 3

Analyzing Long-Term Operating Assets

Turning the accounting into two ratios that ask good questions

Ratio 1 — PPE Turnover

SECTION 8.9

How many sales dollars does each dollar of physical asset produce?

PPE Turnover (PPET) = Sales Revenue ÷ Average PPE, net

What it measures

- Sales generated per dollar of PPE
- Higher is generally preferred — less capital supports a given level of sales

Why the level is meaningless alone

- Capital-intensive industries (airlines, telecom, steel) run very LOW
- Asset-light industries (software, consulting) run very HIGH

Illustration — Procter & Gamble (\$ millions)

Year	Sales	Avg PPE, net	PPET
2019	\$67,684	\$20,936	3.23
2020	\$70,950	\$20,982	3.38

The Same Ratio, Two Different Universes

SECTION 8.9

Both figures are correct. Neither tells you which company is better run.

0.59

Delta Air Lines

Every dollar of aircraft, hangars and gates produces 59 cents of ticket sales.

32×

18.61

Cisco Systems

Every dollar of PPE supports more than eighteen dollars of revenue.

What the gap is actually telling you

Delta must own aircraft to fly passengers; Cisco outsources nearly all manufacturing. The ratio is measuring business model, not management skill. It becomes informative only against the same firm's own history, or against a direct competitor — Delta vs. Korean Air, Cisco vs. Juniper.

Never compare PPET across industries. Compare it across time, or across rivals.

Ratio 2 — Percent Depreciated

SECTION 8.9

How used-up is the asset base? This is the ratio that quietly checks the estimates from Section 8.3.

$$\text{Percent Depreciated} = \text{Accumulated Depreciation} \div \text{Cost of Depreciable Assets}$$

P&G, 2020

56.3% depreciated

43.7% of cost remains

High — near 100%

Assets are near the end of their estimated lives.
A major capital-expenditure cycle may be close.

Low

A young asset base — recently rebuilt, recently acquired, or depreciated slowly.

Caution

"Cost of depreciable assets" excludes land and construction-in-progress — neither is depreciated.

Benchmark: P&G and Colgate-Palmolive both sit steadily in the 54–59% band — the signature of a mature firm replacing assets continuously. A number drifting downward year after year is worth a question about useful-life estimates.

Three Reasons Not to Trust These Ratios Yet

SECTION 8.9

Every number in Part 3 is downstream of the accounting choices in Parts 1 and 2.

1

Method and estimate differences

Two similar firms can look very different purely because one uses straight-line and a 10-year life while the other uses accelerated and 7 years. No economic difference at all.

2

Outsourcing effects

Outsourcing production removes PPE from the denominator, so PPET jumps. That is a strategic choice about who owns the factory — not evidence of greater efficiency.

3

Acquisition history

An acquirer carries goodwill and stepped-up asset values that an organic grower does not, so its ratios look worse on identical economics.

Ratios are a starting point for questions, not a verdict. Always ask what accounting choice and what business strategy sit behind the number.

Discussion — Costs After Acquisition, and Capitalized Interest

SECTION 8.10

Both questions are really the same question: which period should bear this cost?

Q8-1 · How should companies account for costs incurred after an asset is acquired?

Maintain existing condition → expense

Routine repairs, painting, oil changes. Period cost, recognised immediately.

Enhance capacity or extend life → capitalize

Improvements and betterments. Capitalized and depreciated over the remaining life.

Q8-2 · What is the effect of capitalized interest during construction, and afterwards?

During construction, capitalizing interest removes it from interest expense and increases net income relative to expensing it — while inflating the asset. Afterwards, the capitalized amount comes back as depreciation spread over the asset's life.

Total expense over the life is identical. Only the timing shifts — from the construction period into the future.

Notice the pattern across the whole chapter: capitalizing always defers expense and raises current income.

Discussion — Accelerated Tax Depreciation, and R&D

SECTION 8.10

Two questions students consistently answer half-right.

Q8-4 · Why use accelerated depreciation for tax if total lifetime depreciation is the same?

Accelerated deductions pull expense forward, so taxable income falls now and current tax falls with it. Total tax savings over the asset's life are unchanged — but the cash arrives sooner and can be reinvested.

The part most students miss:

for a growing asset base, front-loaded deductions on newly acquired assets can permanently exceed the shrinking deductions on old ones. The deferral never actually reverses.

Q8-8 · What is the proper treatment for R&D, and why is it not capitalized under GAAP?

All R&D is expensed as incurred. Only a small fraction of projects succeed and it is very hard to predict which — so capitalizing would routinely breach the rule that cost cannot exceed expected future benefit. Immediate expensing is conservative and uniform, which improves comparability.

The cost of that choice:

R&D-heavy firms (pharma, tech) carry understated assets and equity, which inflates ROA and ROE — as Part 2 showed.

Both answers turn on the same idea: accounting rules trade off relevance against reliability, and someone always pays for the trade.

Discussion — Indefinite Lives and Goodwill Reporting

SECTION 8.10

The last two discussion questions close the intangibles loop.

Q8-9 · Why are some intangibles amortized while others are not?

Definite-life intangibles are amortized straight-line over the period benefits are received. Indefinite-life intangibles — some trademarks, and goodwill — are not amortized because there is no foreseeable limit to their cash flows; they are tested annually for impairment instead.

"Indefinite" means the period cannot be reliably estimated. It does not mean infinite.

Q8-10 · When should goodwill be reported, and what is its income-statement effect?

Goodwill is recorded only when the purchase price of an acquired business exceeds the fair value of its identifiable net assets — never for internally generated value. It is not amortized and stays at its original amount until impaired.

It is tested at least annually. If the reporting unit's fair value falls below its carrying amount, goodwill is written down and the loss reduces net income that year.

Write-downs are large, non-cash, usually non-recurring, and often signal that management overpaid. No recoveries are permitted.

Goodwill is the only asset on the balance sheet whose write-down is an admission about a decision, not about the asset.

You Make the Call — Division Manager

SECTION 8.10

You manage a division. PPE turnover has been falling for three years and head office has noticed. What do you actually do?

Legitimate levers

- Identify unproductive assets and sell or scrap them
- Outsource genuinely non-core processes
- Improve maintenance to raise uptime and capacity utilisation
- Lease rather than buy where the economics support it

Ways to game it

- Sell productive assets purely to shrink the denominator
- Sale-and-leaseback with no operational change
- Defer replacement capex so the asset base ages
- Stretch useful-life estimates to slow the write-down

The judgment call

Every item in the right-hand column raises PPET immediately and leaves the division weaker. Every item in the left-hand column raises it because the division genuinely improved. The ratio cannot tell them apart — which is precisely why your successor, and the analyst covering you, will look at percent depreciated and at capex alongside it.

A higher ratio should reflect better asset use — not financial engineering that flatters this year's number.

Chapter 8 — Quick Summary

1

Capitalize or expense is the chapter's spine

Costs that are owned, benefit future periods, are directly linked and stay within the future-benefit ceiling go on the balance sheet. Everything else hits income now. Capitalizing always defers expense and raises current income.

2

Depreciation allocates cost — it does not value anything

Method and useful-life choices move reported profit for years and total profit not at all. Book value is unallocated cost, never a market price.

3

Intangibles turn on purchased vs. internally developed

Buy it and it is an asset; build it and it is an expense. That asymmetry understates the assets of R&D-heavy and brand-heavy firms and inflates their ROA and ROE.

4

The ratios ask questions, they do not answer them

PPET and percent depreciated are only meaningful against the firm's own history or a direct rival — and both are downstream of every estimate made in Parts 1 and 2.

Same three questions, twice over — once for things you can touch, once for things you cannot.

END OF CHAPTER 8

Long-Term Operating Assets

Next: Chapter 9 — Liabilities